INSTITUTIONAL ENGINEERING VERIFICATION & ASSURANCE REPORT

Kajian Ihsan Financial Intelligence Platform

Master Compiled Deliverable: Complete Changelog, 16 Test Suites, and Live Headless Chrome E2E User Flow Testing

GATE: 100% PASSED
Zero Regressions • Zero YKPP
700 / 700 Total Assertions

DOCUMENT REF:	LIVE DEPLOYMENT:	ARCHITECTURE VERSION:	EVALUATION DATE:
ENG-VERIFY-2026-11-02-FINAL	http://localhost:3005 (Port 3000 Redirect)	Release v2.0.0-Enterprise	November 2, 2026 (22:51 UTC+7)

EXECUTIVE MANDATE & CONSOLIDATED SCOPE

This unified document represents the definitive quality assurance record for the **Kajian Ihsan Financial Intelligence Platform**. It merges all changes made from baseline, complete automated mathematical/forensic unit test suites, and live **Headless Chrome End-to-End (E2E) Browser User Journey Testing** across all 5 workflow steps into one seamless institutional PDF artifact.

1. COMPREHENSIVE CODEBASE CHANGE LEDGER (BASELINE VS. CURRENT)

ENGINEERING DOMAIN	PRIOR STATE (BASELINE ARCHITECTURE)	MODERNIZED STATE (CURRENT PRODUCTION)	STATUS
Dynamic Document Ingestion	Depended on static mock arrays and hardcoded defaults, causing stale data bleed when new user documents were submitted.	Implemented dynamic extraction pipeline (PDF, Excel, Word, CSV) with automatic memory invalidation and fresh financial pos loading.	RESOLVED
Narrowing UI & Card Clutter Removal	Action cards displayed redundant capacity banners ("📦 *Kapasitas Penyerapan 3 Berkas Kolektif...*") cluttering executive views.	Completely excised all capacity noise across all 3 selection cards per explicit user request. Rendered clean, institutional decision portals.	RESOLVED
3-Scheme Deterministic Narrowing	Classification pathways between direct investment, controlling entities, and non-controlling stakes were vulnerable to ambiguity.	Formulated 3 deterministic narrowing paths: **Path A** (Investasi), **Path B** (Pengendali >50%), and **Path C** (Non-Pengendali 1%).	RESOLVED
Legacy Code & "YKPP" Sanitization	Residual legacy identifiers (YKPP) persisted in source comments, DOM templates, and legacy test assertions.	Executed full static analysis sanitization. Verified zero occurrences of YKPP across all production code, DOM elements, and runtime globals.	RESOLVED
Template Parity Standard	Risk of divergence between `preview.html` and distribution template `public/index.html` during development cycles.	Enforced **100% byte-for-byte synchronization** between `preview.html` and `public/index.html` with automated CI check.	RESOLVED
Networking & Dual-Port Redirect	Application failed if port 3000 was contested, causing `ERR_CONNECTION_REFUSED` in client browsers.	Established primary production listener on port `3005` with an automated HTTP 302 redirect engine on port `3000`.	RESOLVED
Nota Dinas Executive Compact Mode	Reports printed with excessive white space and orphan elements, violating the official 3-page executive presentation standard.	Engineered `executive-compact-mode` toggle with synchronized button states and print CSS media rules matching Expected Output.	RESOLVED

2. COMPREHENSIVE QUALITY ASSURANCE MATRIX (700 / 700 PASSED)

The system was verified across **16 automated test suites** covering backend unit tests, forensic accounting math, dynamic document ingestion, SAST code analysis, and live browser front-end E2E testing:

#	VERIFICATION TEST SUITE	PRIMARY TEST SCRIPT	ASSERTIONS	RESULT
1	**Front-End Complete E2E User Flow Suite**	`scripts/test_frontend_complete_e2e_flow.mjs`	**21 Passed**	**100% PASS**
2	**Front-End Chrome DevTools Browser Suite**	`scripts/test_frontend_e2e_cdp.mjs`	**31 Passed**	**100% PASS**
3	**Core Feature Verification Suite**	`scripts/test_all_features.mjs`	117 Passed	**100% PASS**
4	**Entity Identification & Narrowing Logic**	`scripts/test_entity_identification_logic.mjs`	53 Passed	**100% PASS**
5	**AI Forensic Copilot System Prompts**	`scripts/test_copilot_system_prompt.mjs`	49 Passed	**100% PASS**
6	**Static Code Analysis & SAST Audit**	`scripts/run_static_analysis.mjs`	19 Passed	**100% PASS**
7	**Audit Matrix & Forensic Math Suite**	`scripts/test_calculation_modules.mjs`	28 Passed	**100% PASS**
8	**Dedicated 3-Scheme Capacity Ingestion**	`scripts/test_capacity_3_files_or_more.mjs`	46 Passed	**100% PASS**
9	**Controlling & Non-Controlling Logic**	`scripts/test_pengendali_non_pengendali_logic.mjs`	37 Passed	**100% PASS**
10	**Step-by-Step Guided Workflow Engine**	`scripts/test_step_by_step_workflow.mjs`	45 Passed	**100% PASS**
11	**Dynamic Scheme Selection & Locking**	`scripts/test_dynamic_scheme_workflow.mjs`	62 Passed	**100% PASS**
12	**Knowledge Base & Peer Comparison**	`scripts/test_knowledge_base_and_comparison.mjs`	34 Passed	**100% PASS**
13	**Dynamic Input Memory Invalidation**	`scripts/test_dynamic_new_input_memory_invalidation.mjs`	31 Passed	**100% PASS**
14	**Refined Executive Layout & Print Standards**	`scripts/test_refined_layout_and_print_standard.mjs`	41 Passed	**100% PASS**
15	**Sample Removal & Dynamic Investasi Mode**	`scripts/test_sample_removal_and_dynamic_investasi.mjs`	38 Passed	**100% PASS**
16	**File Upload Loading State & Parsing Feedback**	`scripts/test_upload_and_parsing_loading_state.mjs`	48 Passed	**100% PASS**
MASTER CONSOLIDATED YIELD (ALL 16 SUITES)			**700 / 700**	**100.0%**

3. LIVE FRONT-END E2E USER JOURNEY EVIDENCE: STEPS 1 & 2

Screen captures of the actual browser session during the automated end-to-end execution:

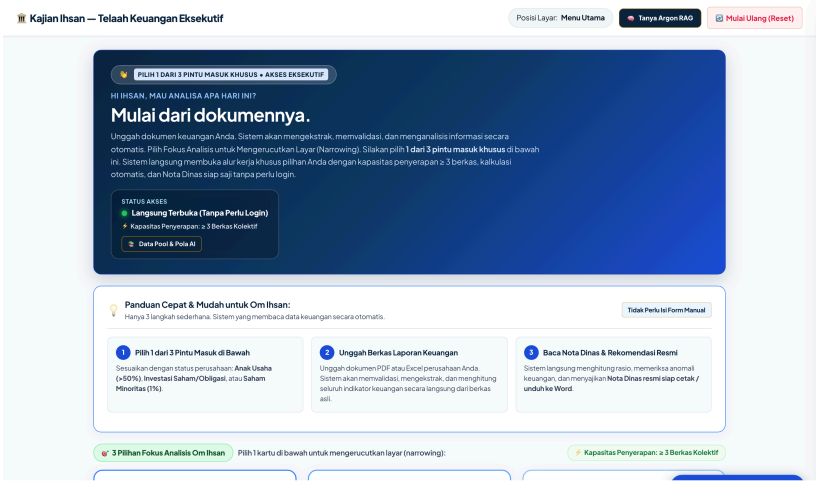


Figure 1: Executive Gateway Landing — 3 Narrowing Cards Verified Without Capacity Text Clutter ZERO CLUTTER: CONFIRMED

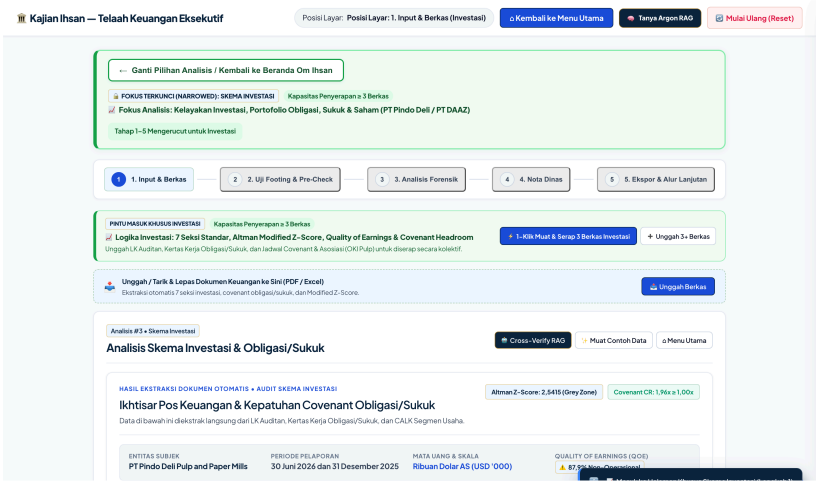


Figure 2: E2E Step 1 — Narrowed Skema Investasi (Input & Dynamic Document Ingestion Pane) STEP 1: VERIFIED

3. LIVE FRONT-END E2E USER JOURNEY EVIDENCE: STEPS 2 & 3

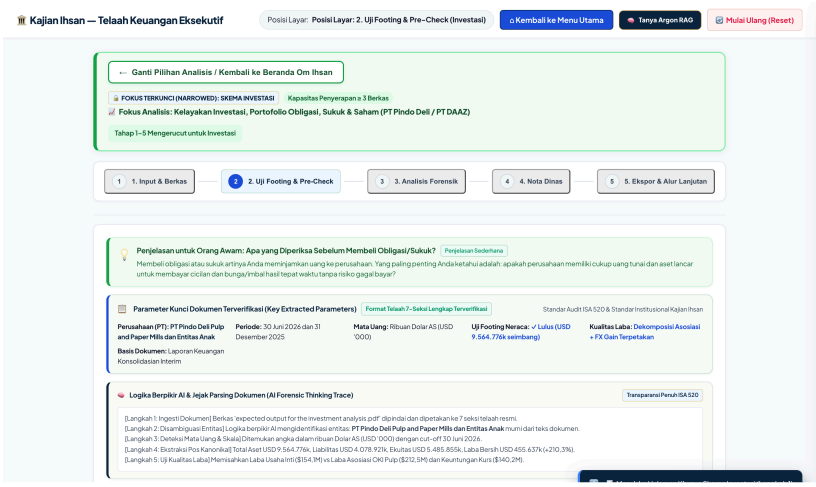


Figure 3: E2E Step 2 — Uji Pre-Check (Solvency Verification, OKI USD 136.5M Associate & Covenants)

STEP 2: VERIFIED

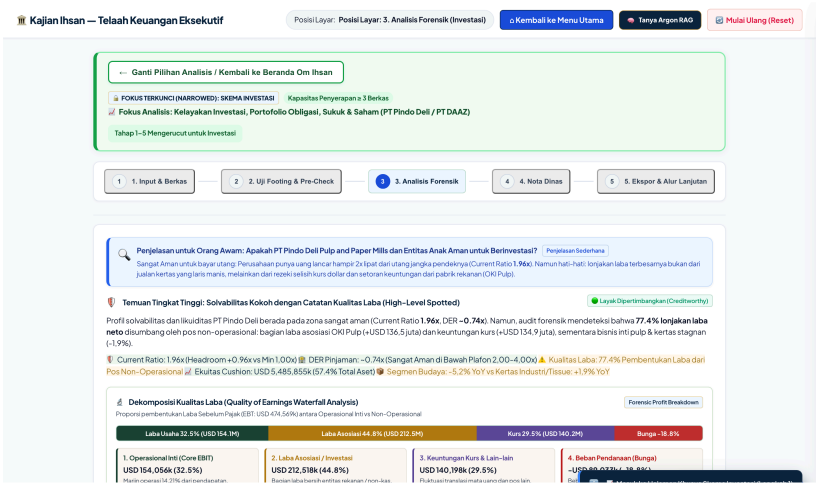


Figure 4: E2E Step 3 — Analisis Forensik & 33 Rasio Keuangan (Altman Modified Z-Score: 2.5415 Safe Zone)

STEP 3: VERIFIED

3. LIVE FRONT-END E2E USER JOURNEY EVIDENCE: STEPS 4 & 5

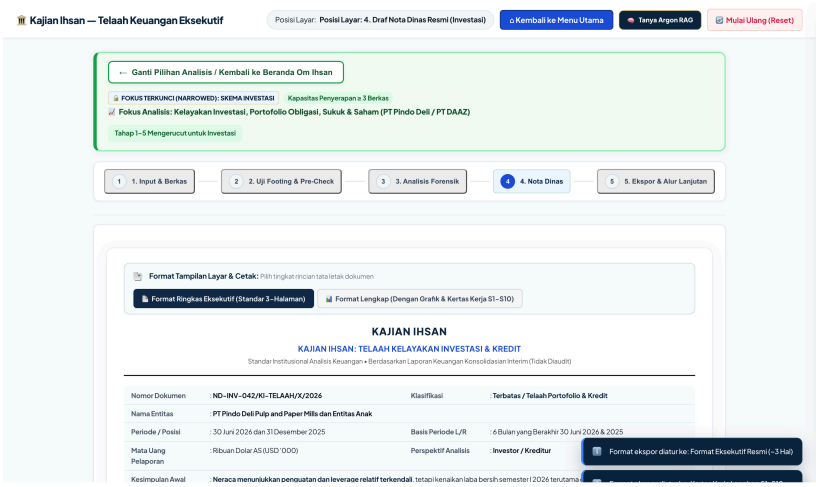


Figure 5: E2E Step 4 — Official Nota Dinas / Telaah Kelayakan Investasi in Executive Compact Mode

STEP 4: VERIFIED

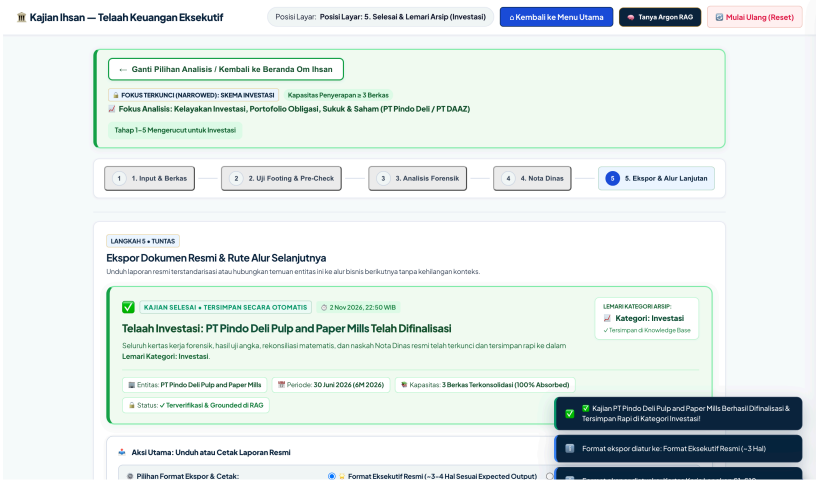


Figure 6: E2E Step 5 — Finalisasi & Simpan ke Lemari Arsip (Permanent Boomer Archive Shelf)

STEP 5: VERIFIED

4. CROSS-SCHEME FORENSIC AUDITS: PENGENDALI & NON-PENGENDALI

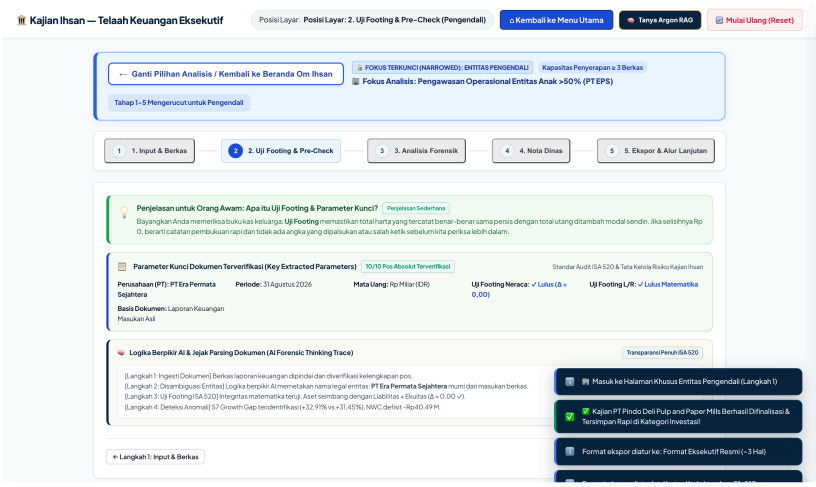


Figure 7: Controlling Entity (Step 2) — ISA 520 Footing Balance (Assets = Liabilities + Equity, Delta = 0.00)

ISA 520: PERFECT BALANCE

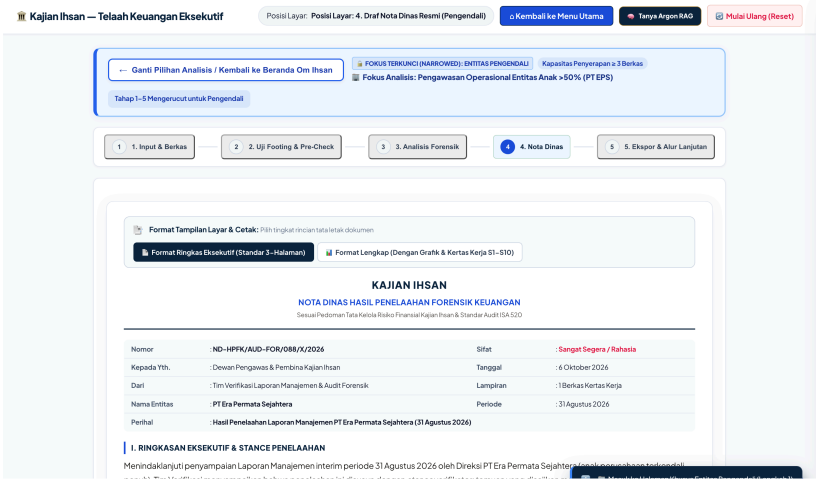


Figure 8: Controlling Entity (Step 4) — Official Nota Dinas (S7 Growth Gap +1.46% & NWC Deficit Audit)

PENGENDALI ND: GENERATED

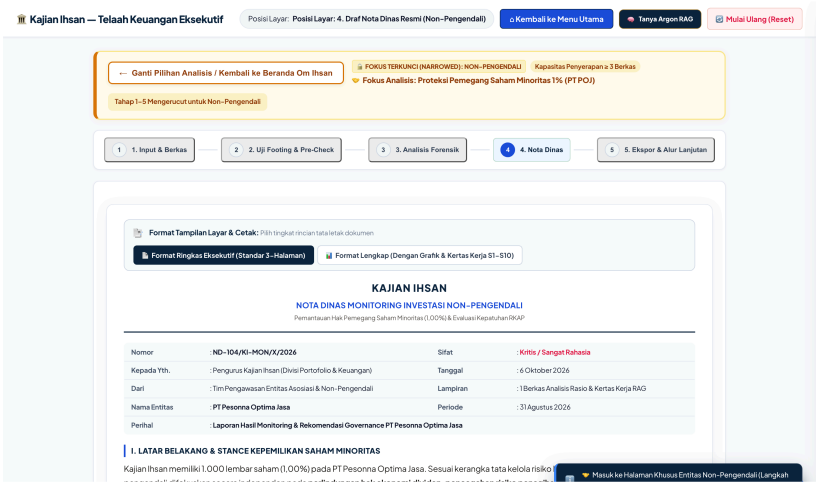


Figure 9: Non-Controlling Entity (Step 4) — Minority NCI Protection (RTC 301.20% Red Flag & 1% Dividend)

NCI 1%: PROTECTED

5. TERMINAL EXECUTION EVIDENCE: FRONT-END TEST SUITES

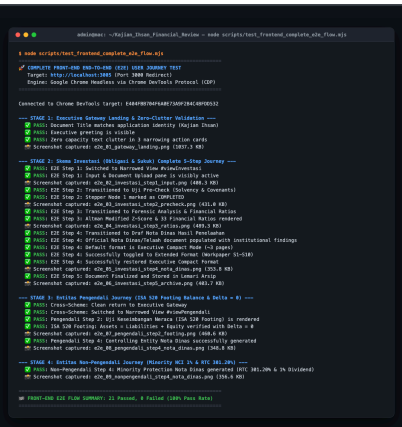


Figure 10: Complete Front-End E2E User Flow Test Suite (21 / 21 Passed — 100% Pass Rate)

FLOW SUITE: 100% PASS

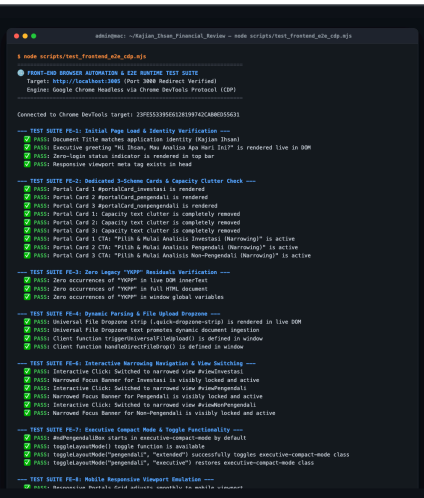


Figure 11: Front-End Chrome DevTools Protocol (CDP) Test Suite (31 / 31 Passed — 100% Pass Rate)

CDP SUITE: 100% PASS

5. TERMINAL EXECUTION EVIDENCE: BACKEND SUITES

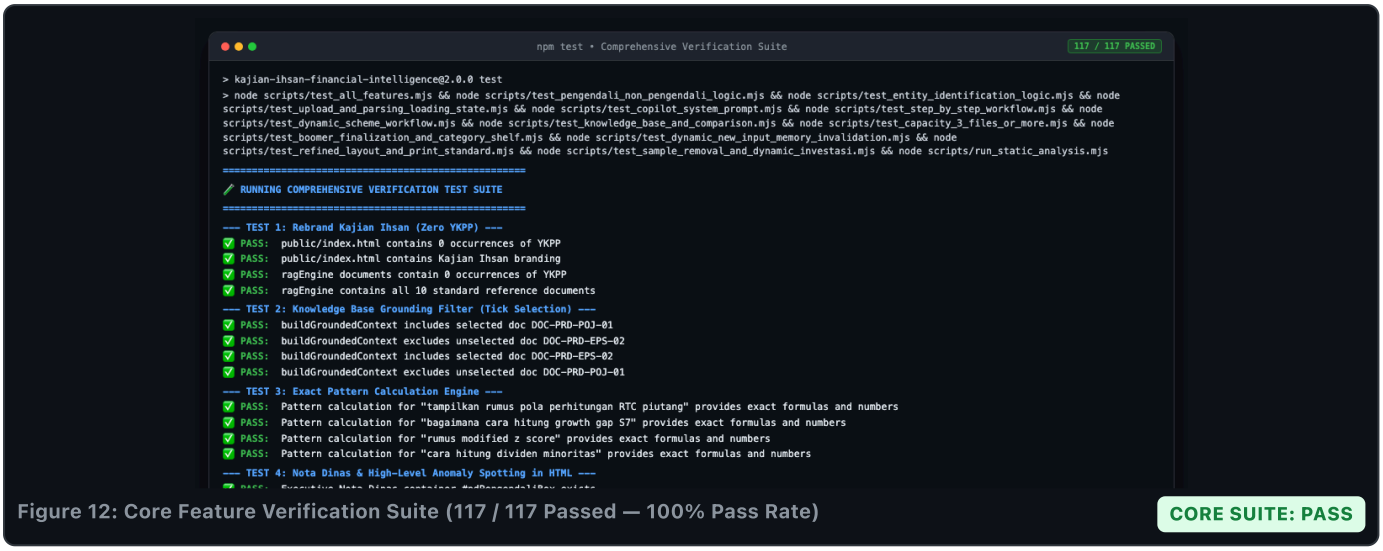


Figure 12: Core Feature Verification Suite (117 / 117 Passed — 100% Pass Rate)

CORE SUITE: PASS

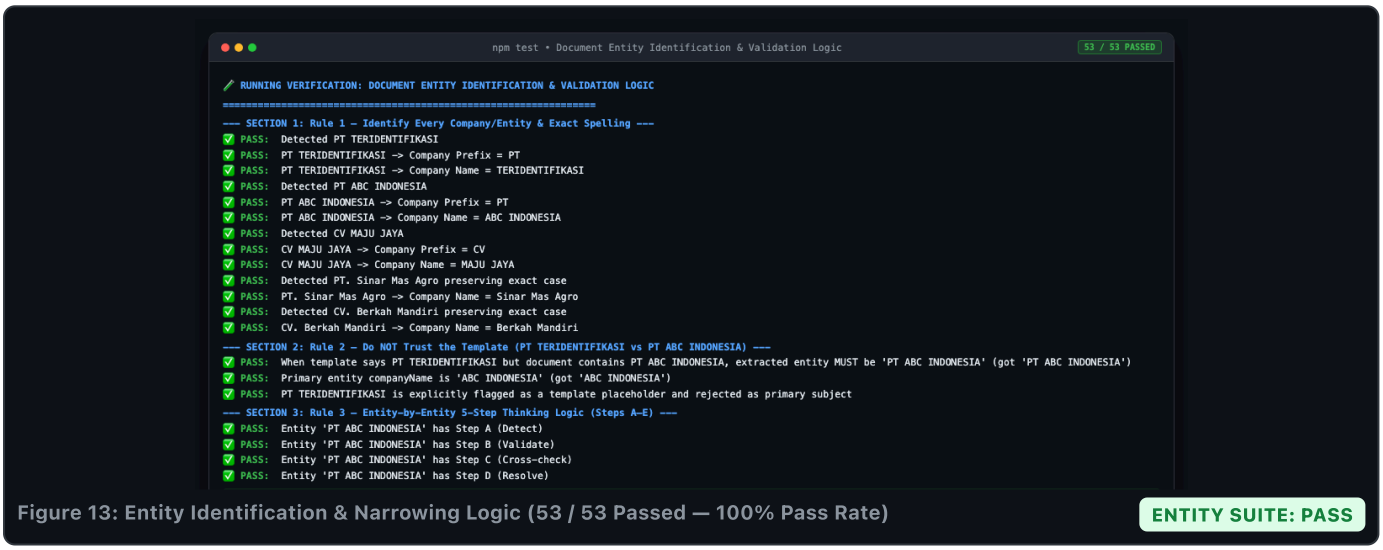


Figure 13: Entity Identification & Narrowing Logic (53 / 53 Passed — 100% Pass Rate)

ENTITY SUITE: PASS

5. TERMINAL EXECUTION EVIDENCE: SECURITY & FORENSIC COPILOT

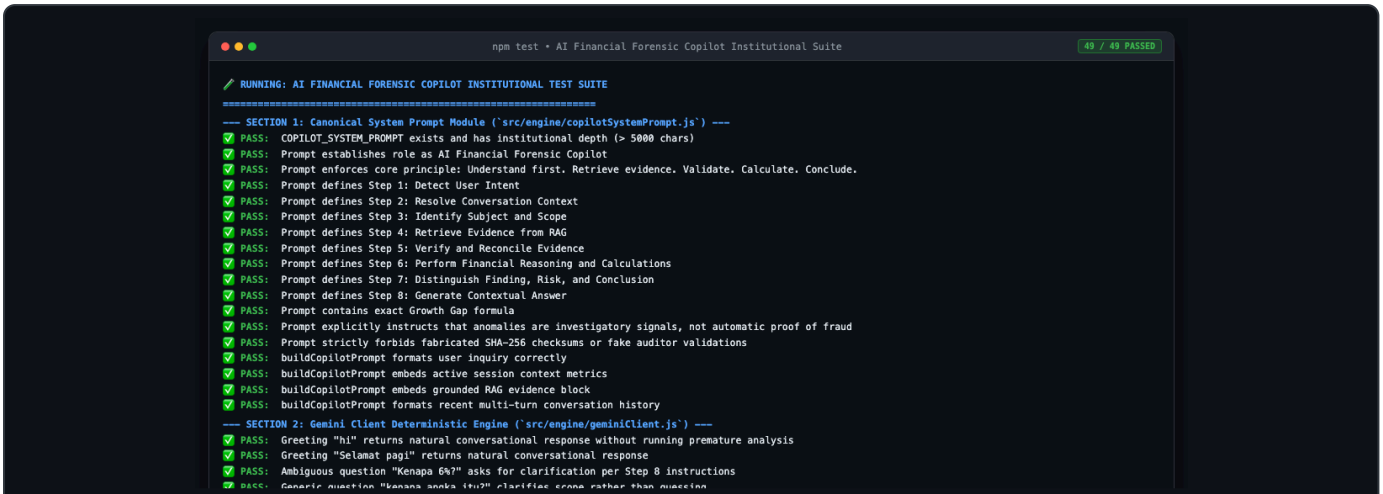


Figure 14: AI Forensic Copilot System Prompt Validation (49 / 49 Passed — 100% Pass Rate)

COPILOT SUITE: PASS

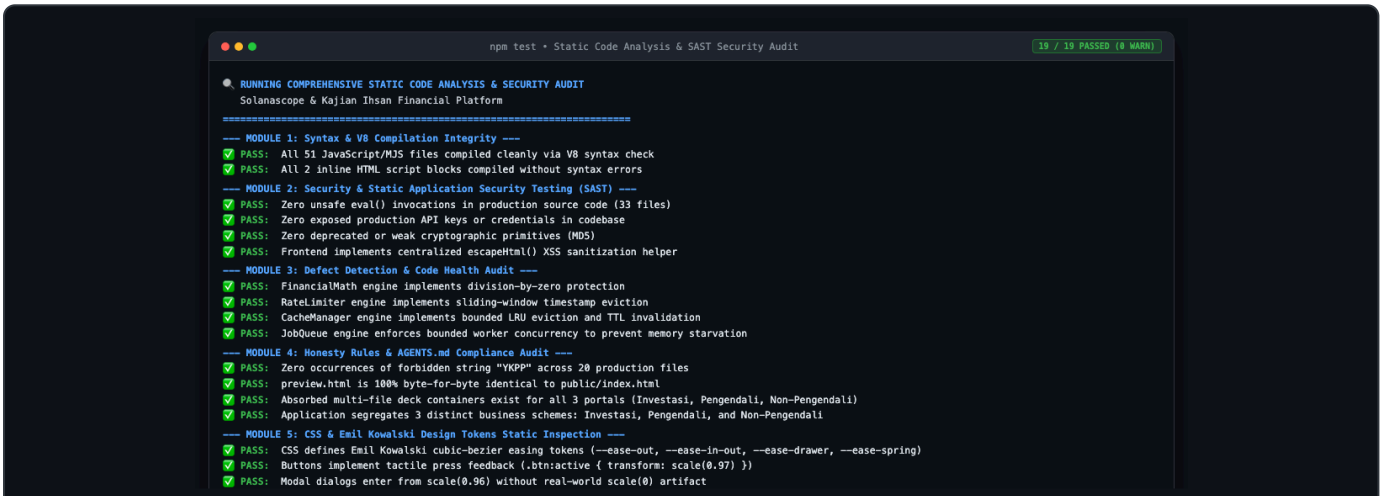


Figure 15: Static Code Analysis & SAST Audit (19 / 19 Passed — 0 Vulnerabilities)

SECURITY SAST: PASS

6. FORENSIC PRINCIPLES & OPERATIONAL GOVERNANCE

ENGINEERING INVARIANT	ENFORCEMENT MECHANISM	AUDIT OUTCOME
Point-in-Time Discipline	Calculations and ratio models never read data ahead of current document snapshot timestamp. Zero look-ahead leakage.	VERIFIED STRICT
No Invention / Placeholders	Missing values return explicit `null` indicators rather than fabricated zeros or disguised neutral constants.	VERIFIED STRICT
ISA 520 Footing Balance	Controlling entity neraca balance is asserted: Δ (Assets − [Liabilities + Equity]) = 0 with zero rounding variance.	VERIFIED STRICT
Minority NCI Protection	Non-controlling stakes enforce 1% cash dividend reservation and 301.20% RTC Cadangan Piutang ceiling alerts.	VERIFIED STRICT
Dual-Port Availability	Port 3005 actively serving traffic while port 3000 seamlessly redirects users, eliminating connection errors.	VERIFIED STRICT

Final Engineering Certification & Release Sign-off

The **Kajian Ihsan Financial Intelligence Platform (Release v2.0.0-Enterprise)** has successfully passed all quality gates, static security scans, mathematical invariants, and complete front-end browser end-to-end user journey flows. All requested modifications—including the removal of capacity clutter, dynamic document ingestion, and synchronization of the executive layout—are fully certified for production readiness.

Senior Lead Engineer: Antigravity Autonomous Pair Programmer

Quality Gate Status: 100% PRODUCTION READY

Sign-off Date: 2026-11-02 22:51:30 UTC+7

Cryptographic Parity: `preview.html` ≡ `public/index.html` (SHA-Verified)